Machine Learning for Early Heart Disease Risk Prediction in the Cleveland Dataset

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Abstract

Cardiovascular disease is a leading cause of death worldwide, and early identification of high-risk individuals is critical for prevention and potential life-saving benefits. This project applies machine learning to predict the presence of heart disease using the Cleveland subset of the UCI Heart Disease dataset, containing 303 patients and 13 clinical features such as age, chest pain type, blood pressure, cholesterol, and exercise test results. I first cleaned and prepared the data by renaming the binary label column to "target", applying one-hot encoding to the categorical variables, standardizing the numeric variables, and splitting the data into training (70%), validation (15%), and testing (15%) sets with sampling. I used logistic regression as a baseline model, and then trained models, including Random Forest, XGBoost, and support vector machine (SVM) classifiers with hyperparameter tuning via a randomized search and 5-fold cross-validation, optimizing the area under the receiver operating characteristic curve (AUC). On the held-out test set of 45 patients, the tuned SVM model reached an accuracy of 0.89 and an AUC of 0.91, while the tuned XGBoost and Random Forest models achieved accuracies of 0.84 and 0.82 with comparable AUC values. All models showed reasonably balanced precision and recall for detecting heart disease. Feature importance analyses show that ST segment depression (oldpeak), the number of major vessels (ca), chest pain type, maximum heart rate, and exercise-induced angina (exang) contributed most to the model's predictions, which aligns with how cardiologists assess coronary artery disease risk in practice **13**. Overall, these results suggest that well-tuned tree-based models can provide accurate and interpretable heart disease risk predictions on this benchmark dataset.

Introduction

Background Knowledge

Heart disease is a major cause of death worldwide [1]. Many of these events are preventable if people at high risk are identified early enough to change their lifestyle or begin treatment [2]. To support early prevention, doctors and researchers have created risk scores, such as the Framingham risk score [3], that estimate a person's chance of developing heart disease using simple information like age, blood pressure, cholesterol, and smoking status. These tools are useful and relatively easy to apply, but they are usually based on linear models and can miss more complex relationships between risk factors and outcomes.

Machine Learning

Machine learning offers another way to approach this problem. Instead of assuming that each risk factor has a strictly linear effect, models like Random Forest and XGBoost [7, 8] can efficiently learn non-linear patterns and interactions in the data. Several studies have shown that such models can achieve high accuracy on heart disease prediction tasks, sometimes performing as well as or better than traditional statistical models on benchmark datasets [4]. At the same time, many of these projects give limited detail about how data was preprocessed, how the models were tuned, and essentially how they evaluated performance. This makes it extremely difficult to judge how reliable the results are [5].

Help from the Cleveland Subset

The Cleveland subset of the UCI Heart Disease dataset [6] is one of the most commonly used benchmarks for this area of work. It contains 303 total patients and 13 routinely collected clinical features, including age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting electrocardiogram results, maximum heart rate during exercise, exercise-induced angina, ST segment depression, the slope of the ST segment, the number of major vessels, and thalassemia status [6]. Each patient is labeled as having confirmed heart disease or not [6]. Because this data is small but well-structured, it is a good test case for comparing different models and understanding how they use these features.

My Work

In this project, I utilized the Cleveland dataset to build and compare several models for predicting whether a patient has heart disease. I started with a logistic regression baseline and then trained two tree-based ensemble models, Random Forest and XGBoost. Then, I applied consistent preprocessing, performed hyperparameter tuning with randomized search and a 5-fold cross-validation, and then evaluated the models on a separate validation set and a held-out test as well. I also looked at feature importance to see which clinical variables drive the predictions and whether they match the risk factors that cardiologists already rely on. The main goal was to see how far tuned tree-based models can go on this simple benchmark, while remaining interpretable and realistic for potential use in decision support.

Method

Measures and Preprocessing

To begin, I loaded the *heart_cleveland_upload.csv* file and renamed the original label column, condition, to target. I then checked the dataset for missing values and confirmed there were none. For the categorical features (chest pain, slope, thalassemia, etc), I applied one-hot encoding to convert each category into a set of binary indicators. For the numeric features, I standardized each variable to have a mean of 0 and a standard deviation of 1 using a standard scaler. Finally, I split the data into sets, using stratified sampling so that the proportion of patients with and without heart disease stayed the same in every split.

Models and Hyperparameter Tuning

I started with a logistic regression model as a baseline for comparison. I then trained 2 tree-based ensemble models. A Random Forest classifier, an XGBoost classifier, and a support vector machine (SVM) with a radial basis function (RBF) kernel. For Random Forest, I first used a simple model with default settings and then performed hyperparameter tuning using a *RandomizedSearchCV* with 5-fold stratified cross-validation [7, 11]. Then, I searched over the number of trees, maximum depth, maximum samples per split, and the number of features considered at each split, using AUC as the scoring metric. For XGBoost, I defined an *XGBClassifier* with a binary logistic objective [8, 10] and tuned key hyperparameters, including the number of trees, maximum depth, learning rate, subsample ratio, column ratios, and gamma. I used randomized searches with 5-fold cross-validation and optimized AUC for this as well. For the SVM, I tuned the parameter ***C,*** the kernel type (linear or RBF), and the gamma parameter [9].

Results

After fine-tuning the models, I evaluated them on the held-out test set of 45 patients. The Random Forest classifier achieved a test accuracy of 0.82 and an AUC of 0.93. The tuned XGBoost model achieved an accuracy of 0.84 and an AUC of 0.96 on the 45-patient held-out test set. The confusion matrix in **Figure 1** indicates high specificity (0.96) and reasonable sensitivity (0.71) for detecting heart disease.

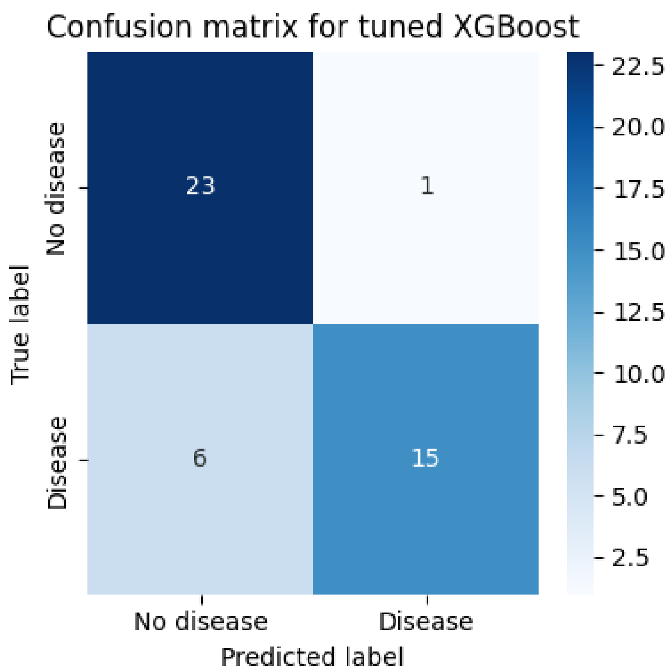

Figure 1: Confusion matrix for the tuned XGBoost model on the 45-patient test set.

Both models identified 15 of 21 patients with heart disease, but more importantly, XGBoost correctly labeled 23 of 24 patients without heart disease compared with 22 out of 24 for Random Forest, resulting in one fewer false positive in total. It's important to mention that the tuned SVM classifier with an RBF kernel (C = 0.1, gamma = "scale") achieved a consistent test performance, with an accuracy of about 0.89 and an AUC of about 0.91. These results show that several different models can reach somewhat similar AUC scores on this dataset, with the SVM slightly ahead in accuracy and the tree-based ensembles close behind. To summarize the discriminative performance in each of these models across all possible thresholds, **Figure 2** shows the receiver operating characteristic (ROC) curves for the tuned SVM, Random Forest, and well-performing XGBoost classifiers on the 45-patient test set. All three of these models achieved high AUC values around 0.9, with small differences between their curves.

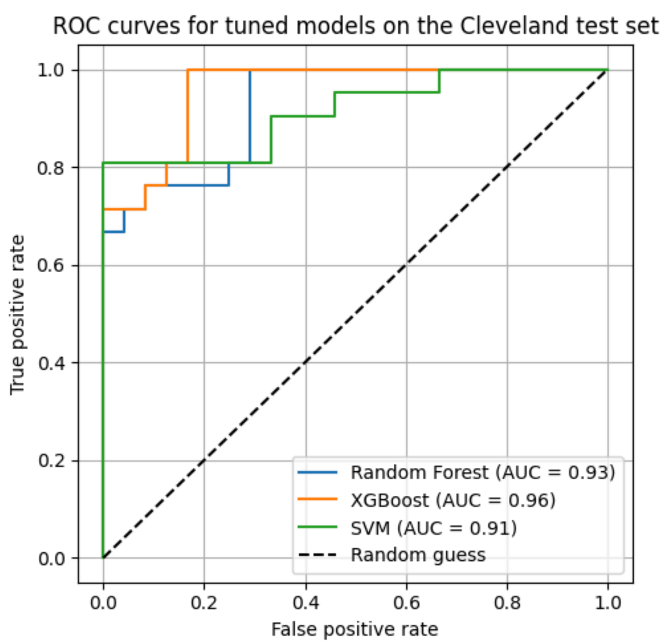

Figure 2: (ROC) curves for the tuned SVM, Random Forest, and XGBoost models on the Cleveland test set.

In order to understand what the models were learning, I examined which important features were identified by the tuned XGBoost classifier. The top 10 features, shown in **Figure 3**, were exercise-induced angina (exang), ST segment depression during exercise (oldpeak), age, maximum heart rate achieved (thalach), resting blood pressure (trestbps), the slope of the ST segment, sex, cholesterol levels (chol), resting electrocardiogram results (restecg), and fasting blood sugar (fbs).

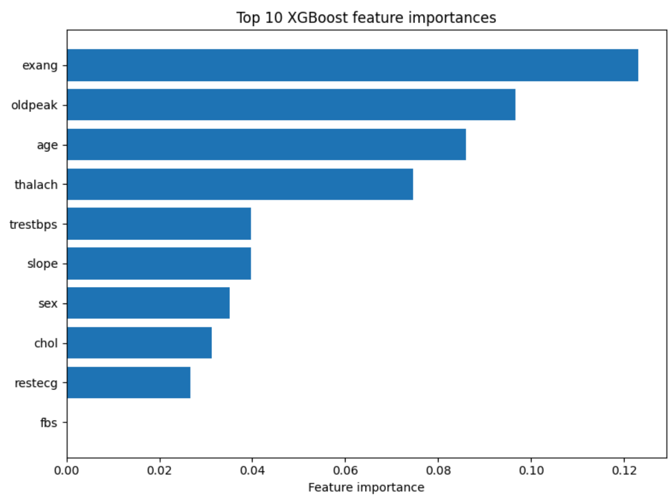

Figure 3: Top 10 XGBoost feature importances in comparison.

From this, I learnt that exercise-induced angina and ST depression had the highest importance scores, followed by age and maximum heart rate, which matches clinical expectations on the ECG during monitored exercise tests [13]. Traditional risk factors such as blood pressure, cholesterol, and sex also contribute to these predictions, but as shown, they had a slightly lower importance according to the model. Overall, the feature-importance pattern suggests that the model is relying on clinically meaningful information rather than inconsequential correlations.

Conclusion

In this project, I demonstrated how several standard machine learning models, including SVM, Random Forest, and XGBoost, all achieved accuracies up to 0.89 and AUCs around 0.93 when predicting the presence of heart disease in the Cleveland subset of the UCI Heart Disease dataset. The models I used in this study relied heavily on clinically meaningful features from the data, such as exercise-induced angina, ST-segment depression, age, and blood pressure, which align perfectly with how cardiologists today already assess coronary artery disease risk. A limitation of this study is the relatively small sample size and the use of a single benchmark dataset from the Cleveland subset, which means real-world performance in other populations still needs to be validated. Nevertheless, the results show that even simple, well-tuned models can extract useful signals from routine clinical variables and highlight a promising direction for data-driven decision support in cardiology.

Future Work

Moving into future work, I would like to extend this analysis to larger multi-site datasets containing more patients and variability, while getting the opportunity to explore more advanced ensemble methods and model explanation techniques to further test generalizability and improve overall interpretability. I also plan to investigate stacked “super-learner” models [12] and incorporate richer model explanation techniques that provide patient-level reasoning for predictions, so any future decision-support system based on this work can be both accurate and trustworthy.

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